

Akshat Gupta — Engineering Knowledge & Experience Portfolio

 **Welcome.** This space is a living knowledge base documenting what I have built, the technologies I have mastered, and the experience I have gained as a software engineer. It is assembled from real evidence — **345 commits and 102 PRs** as the sole engineer on the **Mark AI** signage platform, **244 pull requests** at VibeMonitor, **77 personal repositories**, my résumé, and my weekly 1:1 work log. Use the **Section Directory** below (or the page tree on the left) to navigate.

Akshat Gupta

Software Engineer — Backend · Full-Stack · AI & Observability

B.Tech (CSE – Artificial Intelligence) from KIET (AKTU). Currently a full-time Software Engineer at **Mark AI**, where I am the sole engineer on an AI-native **DOOH** (digital out-of-home) advertising platform — a FastAPI/PostgreSQL signage backend, a Kotlin Android player that runs on the physical screens, a Next.js operator CMS, an advertiser platform, and the AI, 3D and AR layers on top. Previously at **VibeMonitor**, building an AI-driven observability and end-to-end test-automation platform. I own systems end to end — from FastAPI/Python and React/Next.js to AWS, GCP, Kubernetes, CI/CD and LLM-powered features.

At a Glance

Dimension	Detail
Current role	Software Engineer · Mark AI · FULL-TIME
Tenure	Jun 2026 – Present
Mark AI output	102 PRs (100 merged) · 200 commits since 15 Jun 2026 · sole author of all 345 commits across the org's 9 repos
What I own at Mark AI	The whole stack — signage backend (FastAPI + Postgres) · Kotlin player APK · operator CMS · advertiser platform · AI co-pilot · Creative Fit · 3D/AR · GCP infrastructure
Previous role	Software Engineer · VibeMonitor · Sep 2025 – Jun 2026 · 244 PRs (215 merged)
Core repos (VibeMonitor)	vm-api — 119 PRs (Python/FastAPI) · vm-webapp — 118 PRs (Next.js/React)
Primary stack	Python/FastAPI · PostgreSQL · TypeScript/Next.js · Kotlin/Android · AWS & GCP
AI focus	Conversational booking co-pilot (Mark AI) · Agentic RCA & NL→Playwright test generation (VibeMonitor) · RAG · generative creative & 3D
Personal GitHub	77 public repos since 2022 (@akshatgg — github.com/akshatgg)
Roles held	Full-time · Co-founder · Freelancer · Intern → Mentor
Education	B.Tech CSE (AI), KIET (AKTU) · GPA 7.8/10



Section Directory

Open any section to dive in:



Mark AI — What I Built · CURRENT

The current role in depth: 13 themes, 102 PRs, the device contract, proof-of-play, the AI co-pilot and a month-by-month timeline. *Start here for what I'm doing now.*



Knowledge Gained — Languages, Tools & Tech Stack

The full point-wise inventory of every language, framework, tool & methodology — with where and how I applied each.



VibeMonitor — What I Built

A 9-month deep dive on the previous role: 244 PRs across 12 themes, the backend architecture, and a month-by-month timeline.



Personal Projects

Self-initiated builds: Cosmic-Trek, UpInTheSky, Legal E-Library & SipCraft.



Skills Snapshot

Languages — Python · TypeScript · JavaScript · Kotlin · Java · SQL

Backend — FastAPI · SQLAlchemy · Alembic · Node/Express · OAuth2 / JWT · Stripe · Razorpay

Mobile / Signage — Kotlin Android · ExoPlayer · WorkManager · AOSP (no Play Services) · Proof-of-play

Frontend — Next.js 15 · React 19 · Tailwind · Zustand · TanStack Query · Three.js / WebGL

AI / LLM — LangGraph · LangChain · Tool-calling agents · Gemini · RAG · Playwright · BYOLLM · Generative 3D

Data — PostgreSQL · ClickHouse · MongoDB · Redis · Firebase · Cloud SQL

Observability — Prometheus · Grafana · Datadog · New Relic · OpenTelemetry

Cloud / DevOps — AWS · Azure · GCP · Cloud Run (incl. L4 GPU) · Docker · Kubernetes · GitHub Actions

Compiled 4 June 2026 and updated 7 August 2026 — from the Adneuron org (Mark AI: 9 repos, 345 commits, 102 PRs), GitHub (akshatgg + akshatgggg / Vibe-Monitor org), résumé and 1:1 work records. The detailed inventory lives on the Knowledge Gained page.